A. Example of AIME Results

Prompt:

" | User | Let x, y and z be positive real numbers that satisfy the following system of equations:

$$\log_2\left(\frac{x}{yz}\right) = \frac{1}{2}$$

$$\log_2\left(\frac{y}{xz}\right) = \frac{1}{3}$$

$$\log_2\left(\frac{z}{xy}\right) = \frac{1}{4}$$

Then the value of $\left|\log_2(x^4y^3z^2)\right|$ is $\frac{m}{n}$ where m and n are relatively prime positive integers. Find $m+n$. Please reason step by step, and put your final answer within $\square$. | Assistant | "

Our Result:

⟨think⟩...⟨/think⟩ (2275 tokens)

Given the system of equations:

1. $\log_2\left(\frac{x}{yz}\right) = \frac{1}{2}$ 2. $\log_2\left(\frac{y}{xz}\right) = \frac{1}{3}$ 3. $\log_2\left(\frac{z}{xy}\right) = \frac{1}{4}$

We convert these logarithmic equations to their exponential forms:

1. $\frac{x}{yz} = \sqrt{2}$ 2. $\frac{y}{xz} = 2^{1/3}$ 3. $\frac{z}{xy} = 2^{1/4}$

Expressing each variable in terms of x, y, and z:

1. $x = \sqrt{2} \cdot yz$ 2. $y = 2^{1/3} \cdot xz$ 3. $z = 2^{1/4} \cdot xy$

Substituting these expressions into each other, we find:

- From $x = \sqrt{2} \cdot yz$ and $y = 2^{1/3} \cdot xz$, we get $x = \sqrt{2} \cdot (2^{1/3} \cdot xz) \cdot z$, leading to $z = 2^{-5/12}$. - Substituting z into the equation for y, we get $y = 2^{-3/8}$. - Substituting z and y into the equation for x, we get $x = 2^{-7/24}$.

Next, we compute $x^4y^3z^2$:

- $x^4 = (2^{-7/24})^4 = 2^{-7/6}$ - $y^3 = (2^{-3/8})^3 = 2^{-9/8}$ - $z^2 = (2^{-5/12})^2 = 2^{-5/6}$

Multiplying these together:

$$x^4y^3z^2 = 2^{-7/6} \cdot 2^{-9/8} \cdot 2^{-5/6} = 2^{-75/24} = 2^{-25/8}$$

Taking the logarithm base 2:

$$\log_2(x^4y^3z^2) = -25/8$$

The absolute value is:

$$|\log_2(x^4y^3z^2)| = 25/8$$

Thus, $m = 25$ and $n = 8$, giving $m + n = 33$.

$$\boxed{33}$$

Baseline Result:

⟨think⟩...⟨/think⟩ (9392 tokens)

Given the system of logarithmic equations:

$$\log_2\left(\frac{x}{yz}\right) = \frac{1}{2}$$
$$\log_2\left(\frac{y}{xz}\right) = \frac{1}{3}$$
$$\log_2\left(\frac{z}{xy}\right) = \frac{1}{4}$$

We convert these equations to exponential form:

$$\sqrt{2} = \frac{x}{yz} \implies x = \sqrt{2} \cdot yz$$
$$2^{\frac{1}{3}} = \frac{y}{xz} \implies y = 2^{\frac{1}{3}} \cdot xz$$
$$2^{\frac{1}{4}} = \frac{z}{xy} \implies z = 2^{\frac{1}{4}} \cdot xy$$

Substituting these into each other, we find:

1. $x = \sqrt{2} \cdot yz$ 2. $y = 2^{\frac{1}{3}} \cdot xz$ 3. $z = 2^{\frac{1}{4}} \cdot xy$

By solving these equations step-by-step, we find:

- $z = 2^{-5/12}$ - $xy = 2^{-2/3}$ - $y = 2^{-1/24}$ - $x = 2^{1/24}$

However, upon verifying the third equation, we realize the system might not have a solution. Therefore, we use matrix inversion to solve the system:

$$\begin{cases} \log_2(x) - \log_2(y) - \log_2(z) = \frac{1}{2} \\ \log_2(y) - \log_2(x) - \log_2(z) = \frac{1}{3} \\ \log_2(z) - \log_2(x) - \log_2(y) = \frac{1}{4} \end{cases}$$

Let $A = \log_2(x)$, $B = \log_2(y)$, $C = \log_2(z)$. The system becomes:

$$\begin{cases} A - B - C = \frac{1}{2} \\ B - A - C = \frac{1}{3} \\ C - A - B = \frac{1}{4} \end{cases}$$

We solve this system using Cramer's rule and matrix inversion. The determinant of the matrix is -2, and the solutions are:

$$A = \frac{61}{48}, \quad B = \frac{13}{16}, \quad C = -\frac{1}{6}$$

Thus, we have:

$$x = 2^{61/48}, \quad y = 2^{13/16}, \quad z = 2^{-1/6}$$

We need to find $\left|\log_2(x^4y^3z^2)\right|$:

$$4\log_2 x + 3\log_2 y + 2\log_2 z = 4 \cdot \frac{61}{48} + 3 \cdot \frac{13}{16} + 2 \cdot \left(-\frac{1}{6}\right)$$

Converting to a common denominator (48):

$$4 \cdot \frac{61}{48} = \frac{244}{48}, \quad 3 \cdot \frac{13}{16} = \frac{117}{48}, \quad 2 \cdot \left(-\frac{1}{6}\right) = -\frac{16}{48}$$

Summing these:

$$\frac{244}{48} + \frac{117}{48} - \frac{16}{48} = \frac{345}{48} = \frac{115}{16}$$

The fraction $\frac{115}{16}$ is in simplest form, so $m = 115$ and $n = 16$. Therefore, $m + n = 115 + 16 = 131$.

$$\boxed{131}$$

Prompt:

" | User | Real numbers x and y with $x, y > 1$ satisfy $\log_x(y^x) = \log_y(x^{4y}) = 10$. What is the value of xy?
Please reason step by step, and put your final answer within $\square$. | Assistant | "

Our Result:

⟨think⟩...⟨/think⟩ (15147 tokens)

Given the equations:

1. $\log_x(y^x) = 10$
2. $\log_y(x^{4y}) = 10$

We start by applying the logarithm power rule to both equations.

For the first equation:

$$\log_x(y^x) = x \cdot \log_x(y) = 10$$

Let $z = \log_y(x)$. Then, $\log_x(y) = \frac{1}{z}$. Substituting this into the equation, we get:

$$x \cdot \frac{1}{z} = 10 \implies x = 10z$$

For the second equation:

$$\log_y(x^{4y}) = y \cdot \log_y(x^4) = 10$$

Using the power rule, we get:

$$y \cdot 4\log_y(x) = 10 \implies 4y\log_y(x) = 10$$

Since $\log_y(x) = z$, this simplifies to:

$$4yz = 10 \implies yz = \frac{5}{2}$$

We now have two expressions:
1. $x = 10z$
2. $y = \frac{5/2}{z}$

Substituting $x = 10z$ and $y = \frac{5/2}{z}$ into the product xy, we get:

$$xy = (10z) \cdot \left(\frac{5/2}{z}\right) = 10z \cdot \frac{5}{2z} = 10 \cdot \frac{5}{2} = 25$$

Thus, the value of xy is $\boxed{25}$.

Baseline Result:

⟨think⟩...⟨/think⟩ (16223 tokens)

Given the problem where real numbers x and y with $x, y > 1$ satisfy the equations $\log_x(y^x) = 10$ and $\log_y(x^4y) = 10$, we need to find the value of xy.

First, we convert the logarithmic equations to their exponential forms using the change of base formula.